CS130 - Math review

Name: _____ SID: _____

Part 1: vectors, dot and cross product

1. Calculate the cosine of the angle between the vectors $\begin{pmatrix}2\\4\\4\end{pmatrix}$ and $\begin{pmatrix}4\\3\\0\end{pmatrix}$.

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2. Consider a plane containing the point $P = (1, 3, -1)$, with normal $\mathbf{n} = (\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})$. Let $Q = (5, 6, 7)$ be a point (not on the plane). For each point below, indicate whether the point is on the plane, on the same side of the plane as Q, or on the opposite side of the plane from Q.

1. $(0, 0, 0)$
2. $(-1, 1, 2)$
3. $(1, 4, 0)$
4. $(1, 5, 1)$
5. $(-1, -1, -1)$

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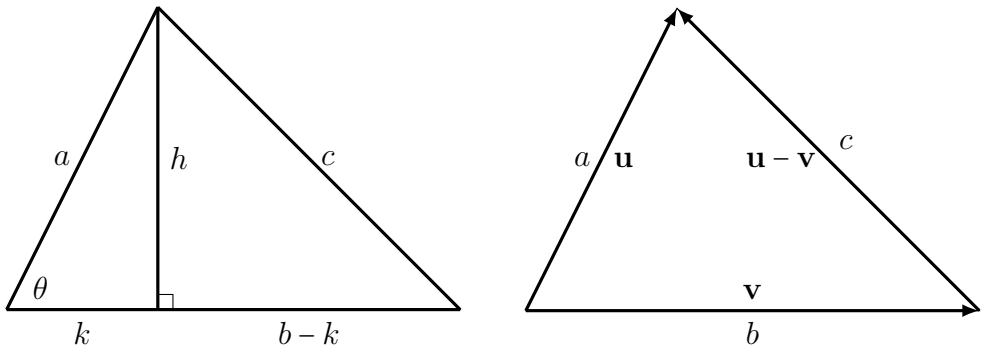
3. Calculate the cross product: $\begin{pmatrix}1\\2\\3\end{pmatrix} \times \begin{pmatrix}4\\5\\6\end{pmatrix}$.

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4. Calculate a vector $\mathbf{w}$ in the same direction of the vector $\mathbf{u}$ and that has the same length as the vector $\mathbf{v}$.

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5. Consider a triangle with sides $\mathbf{u}$ (of length a), $\mathbf{v}$ (of length b), and $\mathbf{u}-\mathbf{v}$ (of length c), with the angle between $\mathbf{u}$ and $\mathbf{v}$ equal to θ as shown in the two figures below.



In parts (a)-(g), you will derive the identity $\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$. In parts (h)-(i), you will derive the identity $\|\mathbf{u}\times\mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin\theta$.

(a) Write an identity that relates a, h, and k.

(b) Write an identity that relates c, h, and $b-k$.

(c) Write an identity that relates a, k, and θ.

(d) From the three identities above, eliminate the variables h and k. This should leave you with the Law of Cosines, $c^2 = a^2 + b^2 - 2ab\cos\theta$.

(e) Rewrite the Law of Cosines in terms of the vectors $\mathbf{u}$, $\mathbf{v}$, and θ. a, b, and c should not appear in your expression.

(f) Expand the left hand side. This should result in a term containing $\mathbf{u}\cdot\mathbf{v}$.

(g) Solve for $\mathbf{u}\cdot\mathbf{v}$ and simplify.

(h) Let the components of $\mathbf{u}$ and $\mathbf{v}$ be given as $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$. Show algebraically, by expanding in terms of the components, that

$$\|\mathbf{u}\times\mathbf{v}\|^2 + (\mathbf{u}\cdot\mathbf{v})^2 = \|\mathbf{u}\|^2\|\mathbf{v}\|^2.$$

(i) Use the identity from part (h) and the expression derived in part (g) to solve for $\|\mathbf{u}\times\mathbf{v}\|$, and simplify.

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6. Given the triangle with vertices $(0,2,-1)$, $(2,0,-1)$, and $(1,0,0)$, calculate the normal of the plane that contains the triangle.

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Part 2: Matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 5 \\ 3 & 7 & 19 \end{pmatrix} \qquad \mathbf{B} = \begin{pmatrix} 5 & 2 \\ 1 & -3 \\ -1 & 1 \end{pmatrix} \qquad \mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

7. Calculate: (a) $\mathbf{A} + \mathbf{B}^T$, (b) $\mathbf{AB}$, and (c) $(\mathbf{AB})^{-1}$.

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8. Solve $(\mathbf{AB})\mathbf{x} = \mathbf{u}$ for $\mathbf{x}$.

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9. Show that $\mathbf{AB} \neq \mathbf{BA}$.

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10. Let $\mathbf{C} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix}$, and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$. Show that $(\mathbf{CD})\mathbf{v} = \mathbf{C}(\mathbf{Dv})$ by calculating $\mathbf{A} = \mathbf{CD}$, $\mathbf{w} = \mathbf{Dv}$, $\mathbf{Av}$, and $\mathbf{Cw}$.

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